Transformation of Variables

1 Model Inadequacies and Transformations

Correcting Model Inadequacies

- Least-squares inference assumes a linear relationship, constant error variance, and normally distributed errors
- Residual plots reveal when these assumptions fail
- One standard remedy is to transform the data; weighting is a separate topic
- In this lecture: transforming y to stabilize the variance, transforming y or x to linearize the model, and two analytical methods for choosing the power (Box–Cox for y, Box–Tidwell for x)

Recognizing Nonconstant Variance

- Plot R-student against the fitted values $\hat{y}_i$
- Outward-opening funnel: the spread grows with $\mathrm{E}[y_i]$, common when y is a count or an amount
- With nonconstant variance the OLS estimators remain unbiased but are no longer minimum-variance, and the usual t and F tests are unreliable

Idea of a Variance-Stabilizing Transformation

- Suppose $\mathrm{Var}(y_i)$ depends on the mean $\mathrm{E}[y_i]$
- Replace y by $y^* = h(y)$, with h chosen so that $\mathrm{Var}(y^*)$ is approximately constant
- Then fit the model on the transformed scale

$$y_i^* = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \mathrm{Var}(\varepsilon_i) \approx \sigma^2$$

- The appropriate h depends on how $\mathrm{Var}(y)$ is related to $\mathrm{E}[y]$

2 Variance-Stabilizing Transformations

Variance-Stabilizing Transformations

Relationship of σ^2 to $\mathrm{E}[y]$	Transformation
$\sigma^2 \propto$ constant	$y^* = y$ (no transformation)
$\sigma^2 \propto \mathrm{E}[y]$	$y^* = \sqrt{y}$ (Poisson data)
$\sigma^2 \propto (\mathrm{E}[y])^2$	$y^* = \ln y$
$\sigma^2 \propto (\mathrm{E}[y])^3$	$y^* = y^{-1/2}$
$\sigma^2 \propto (\mathrm{E}[y])^4$	$y^* = y^{-1}$

Where the Table Comes From: The Delta Method

Let $\mu = \mathrm{E}[y]$. A first-order Taylor expansion of h about μ gives

$$h(y) \approx h(\mu) + h'(\mu)\,(y - \mu) \quad \Longrightarrow \quad \mathrm{Var}\left(h(y)\right) \approx \left[h'(\mu)\right]^2 \mathrm{Var}(y)$$

- $\mathrm{Var}(h(y))$ is approximately constant when $h'(\mu) \propto 1/\sqrt{\mathrm{Var}(y)}$
- So h is found by integrating one over the standard deviation, viewed as a function of the mean

The Delta Method for Power Relationships

Suppose the standard deviation is a power of the mean, $\sigma \propto \mu^a$, that is $\sigma^2 \propto \mu^{2a}$. Then

$$h'(\mu) \propto \mu^{-a} \quad \Longrightarrow \quad h(y) = \begin{cases} y^{1-a}, & a \neq 1 \\ \ln y, & a = 1 \end{cases}$$

- $a = 1/2$: $y^{1/2}$; $a = 1$: $\ln y$; $a = 3/2$: $y^{-1/2}$; $a = 2$: y^{-1}, the rows of the table
- The faster σ grows with μ, the smaller the power $1 - a$, and the stronger the transformation

Reading the Table

- The stronger the dependence of σ^2 on $\mathrm{E}[y]$, the stronger the transformation: $\sqrt{y}$, then $\ln y$, then $y^{-1/2}$, then y^{-1}
- Poisson counts have $\mathrm{Var}(y) = \mathrm{E}[y]$, so $\sigma^2 \propto \mathrm{E}[y]$ and the square root applies
- $\sigma^2 \propto (\mathrm{E}[y])^2$ means the standard deviation is proportional to the mean, which calls for the logarithm

Choosing a Transformation in Practice

1. Fit the model by OLS on the original y and plot R-student against $\hat{y}_i$
2. If the pattern of spread matches a row of the table, apply that transformation
3. Refit with y^* and examine the residual plots again
4. If the variance is still not constant, try a stronger or weaker transformation

The exact relationship between σ^2 and $\mathrm{E}[y]$ is rarely known, so the choice is often made by trial and error.

Consequences of Transforming the Response

- Estimates and tests refer to the y^* scale; for example, coefficients describe the change in $\sqrt{y}$, not in y
- R^2 values from models with different responses cannot be compared; to compare fits on the y scale, use for example the squared correlation between y_i and the back-transformed fitted values $h^{-1}(\widehat{y}_i^*)$
- Transforming y also changes the shape of the relationship and the error distribution, so all residual checks must be repeated

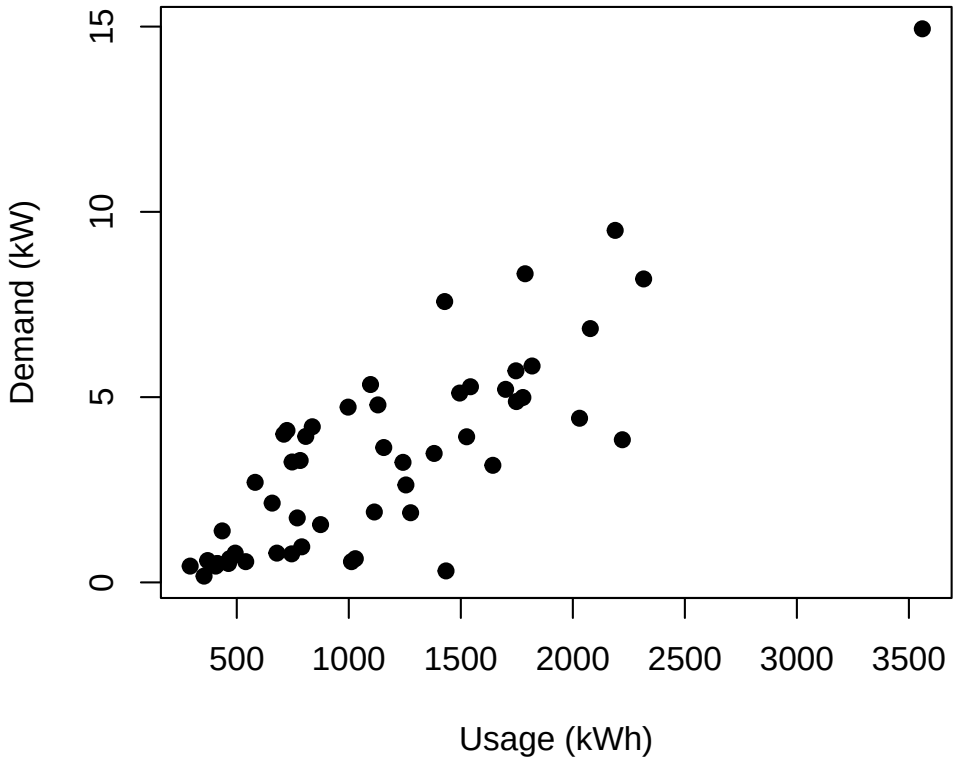
Back-Transforming to the Original Scale

- h is monotone, so it preserves quantiles: a prediction interval (L^*, U^*) for y^* gives the prediction interval $(h^{-1}(L^*),\ h^{-1}(U^*))$ for y (endpoints swap if h is decreasing, as for y^{-1})
- The fitted value $h^{-1}(\widehat{y}^*)$ estimates the *median* of y, not its mean, when the errors on the y^* scale are symmetric; likewise a confidence interval for $\mathrm{E}[y^*]$ back-transforms to one for the median of y
- Log scale, $y^* \sim N(\mu^*, \sigma^2)$: the median of y is e^{μ^*} but $\mathrm{E}[y] = e^{\mu^* + \sigma^2/2}$
- Square-root scale: $y = (y^*)^2$, so $\mathrm{E}[y] = (\mu^*)^2 + \sigma^2$

3 Example: Electric Utility Data

Electric Utility Data

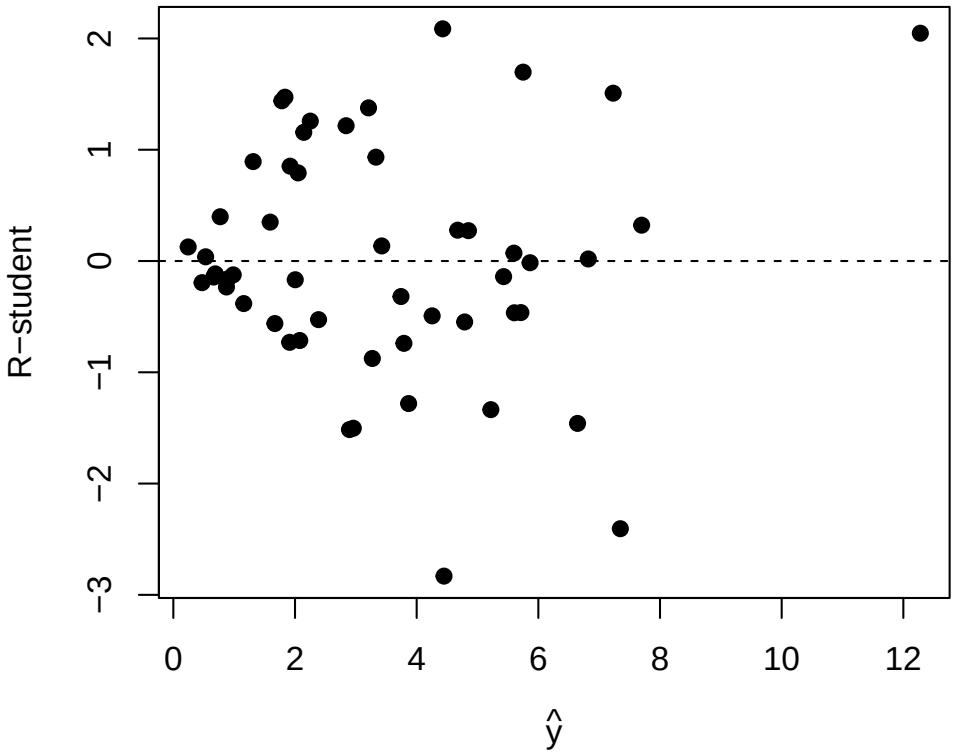
- y: peak demand (kW), x: monthly energy usage (kWh)
- $n = 53$ residential customers in August (MPV Table 5.2)
- Goal: model demand as a function of usage
- The spread of y increases with x



Straight-Line Fit

$$\hat{y} = -0.8313 + 0.003683\,x$$
$$R^2 = 0.7046$$

- R-student against $\hat{y}$ shows an outward-opening funnel
- The constant-variance assumption is violated



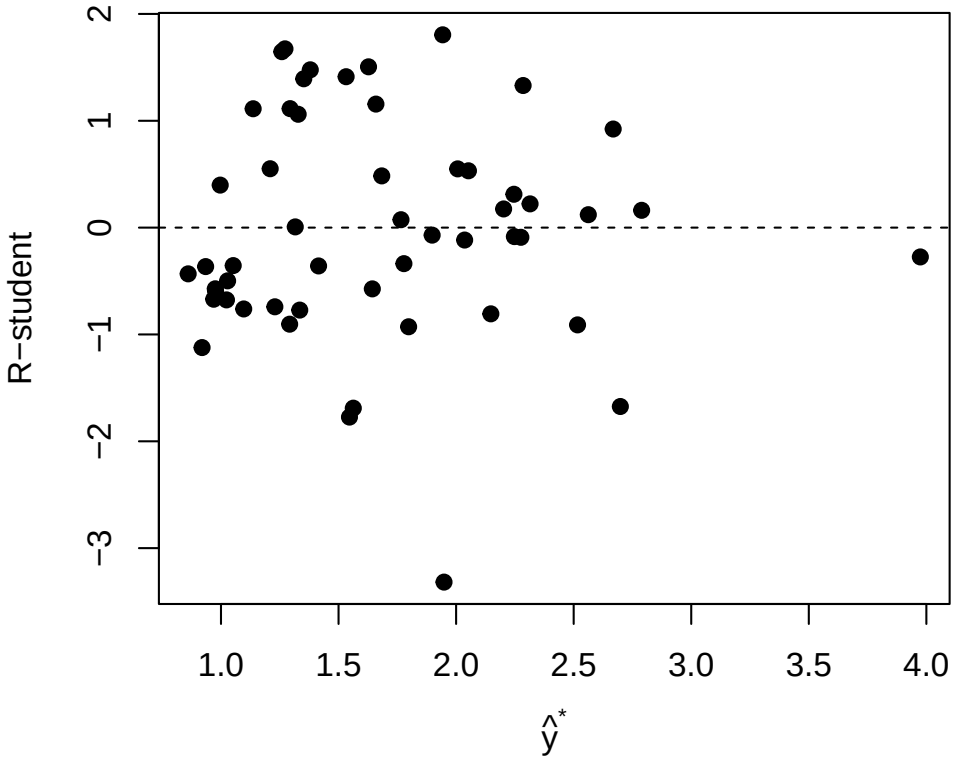
Question: Which Transformation?

The residual plot is an outward-opening funnel, so the variance increases with the mean. Which row of the variance-stabilizing table would you try first, and what would you do if the funnel persisted?

Square-Root Transformation

$$
\begin{aligned}
y^* &= \sqrt{y} \\
\hat{y}^* &= 0.5822 + 0.0009529\, x \\
R^2 &= 0.6485
\end{aligned}
$$

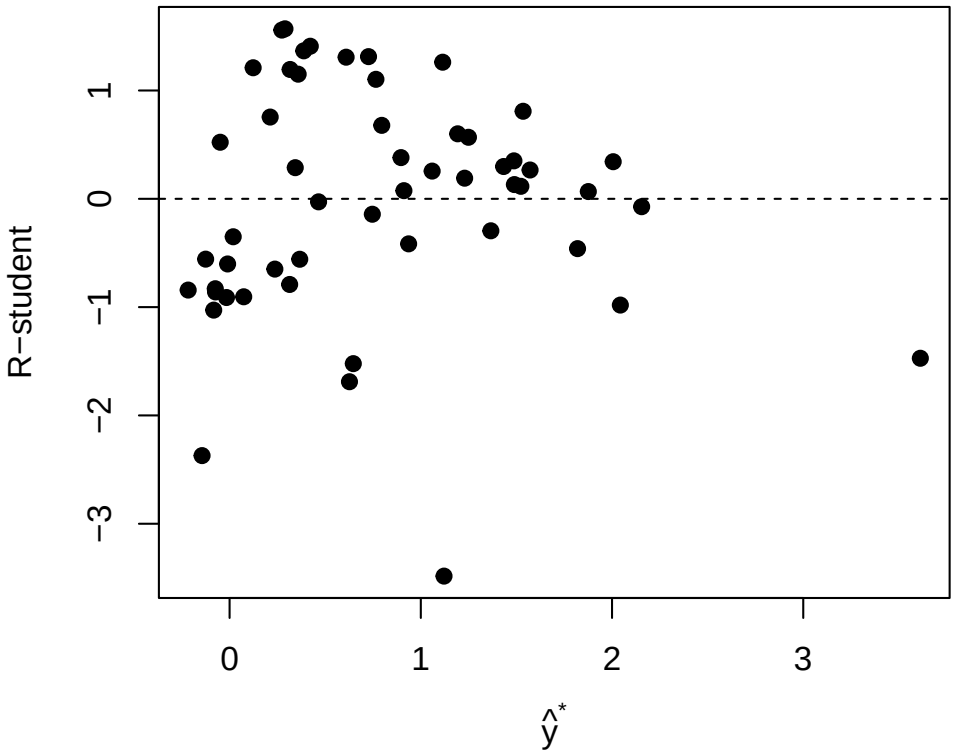
- R-student against $\hat{y}^*$ shows no funnel: the variance now looks stable
- Still check the R-student near -3.3 and the isolated large $\hat{y}^*$, as in the previous lecture



A Stronger Transformation: $\ln y$

$$
\begin{aligned}
y^* &= \ln y \\
\hat{y}^* &= -0.5587 + 0.001172\, x \\
R^2 &= 0.5266
\end{aligned}
$$

- The residual plot now shows curvature
- The logarithm is too strong for these data



Question: Did R^2 Get Worse?

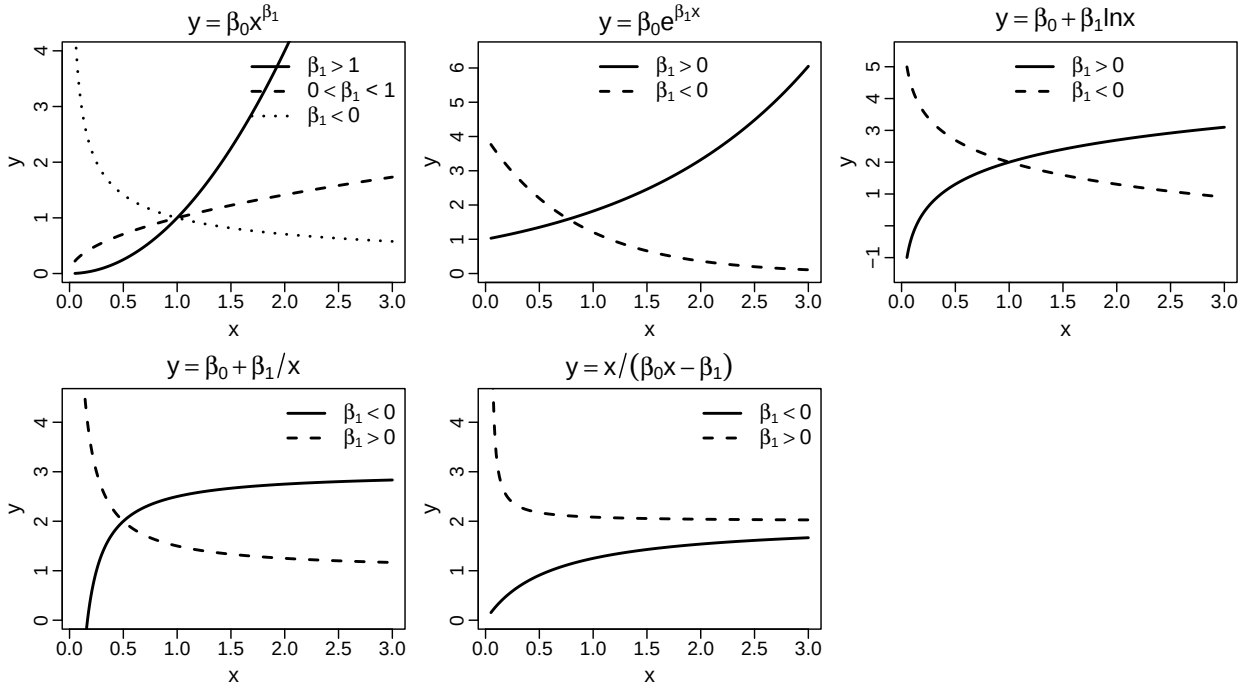
R^2 dropped from 0.7046 for y to 0.6485 for $\sqrt{y}$. Does this mean the transformed model is worse?

4 Transformations to Linearize the Model

Linearizable Nonlinear Functions

Function	Transformation	Linear form
$y = \beta_0 x^{\beta_1}$	$y^* = \ln y,\ x^* = \ln x$	$y^* = \ln \beta_0 + \beta_1 x^*$
$y = \beta_0 e^{\beta_1 x}$	$y^* = \ln y$	$y^* = \ln \beta_0 + \beta_1 x$
$y = \beta_0 + \beta_1 \ln x$	$x^* = \ln x$	$y = \beta_0 + \beta_1 x^*$
$y = \beta_0 + \dfrac{\beta_1}{x}$	$x^* = \dfrac{1}{x}$	$y = \beta_0 + \beta_1 x^*$
$y = \dfrac{x}{\beta_0 x - \beta_1}$	$y^* = \dfrac{1}{y},\ x^* = \dfrac{1}{x}$	$y^* = \beta_0 - \beta_1 x^*$

Shapes of the Linearizable Functions



Match the scatter plot to a panel, then use that row of the table.

The Error Structure Matters

- Linearizing the exponential function assumes the error is multiplicative

$$y = \beta_0 e^{\beta_1 x}\,\varepsilon \quad \Longrightarrow \quad \ln y = \ln \beta_0 + \beta_1 x + \ln \varepsilon$$

- The linearized model is appropriate if $\ln \varepsilon$ has mean zero, constant variance, and is normal
- If instead the error is additive, $y = \beta_0 e^{\beta_1 x} + \varepsilon$, taking logarithms does not give a linear model with additive error, and nonlinear regression is needed

Interpreting Coefficients After a Log Transformation

- $\ln y = \beta_0 + \beta_1 x + \varepsilon$: a unit increase in x multiplies the median of y by e^{β_1}, a change of $100(e^{\beta_1} - 1)\%$, which is about $100\beta_1\%$ when β_1 is small
- $y = \beta_0 + \beta_1 \ln x + \varepsilon$: a 1% increase in x changes $\mathrm{E}[y]$ by $\beta_1 \ln 1.01 \approx \beta_1/100$
- $\ln y = \beta_0 + \beta_1 \ln x + \varepsilon$: a 1% increase in x multiplies the median of y by 1.01^{β_1}, about a $\beta_1\%$ change; β_1 is the elasticity of y with respect to x

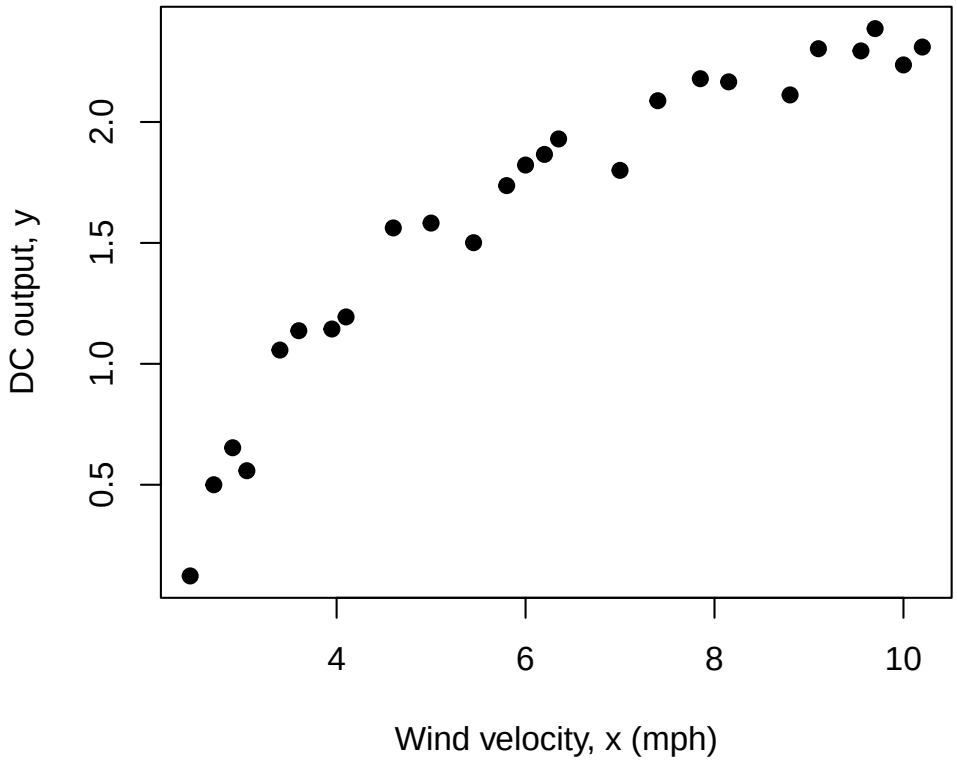
Transform x or Transform y?

- Transforming x alone leaves the distribution of the errors unchanged
- Transforming y changes both the shape of the relationship and the error distribution
- Curved scatter with roughly constant variance: transform x
- Curved scatter with nonconstant variance: transform y, which may fix both problems

5 Example: Windmill Data

Windmill Data

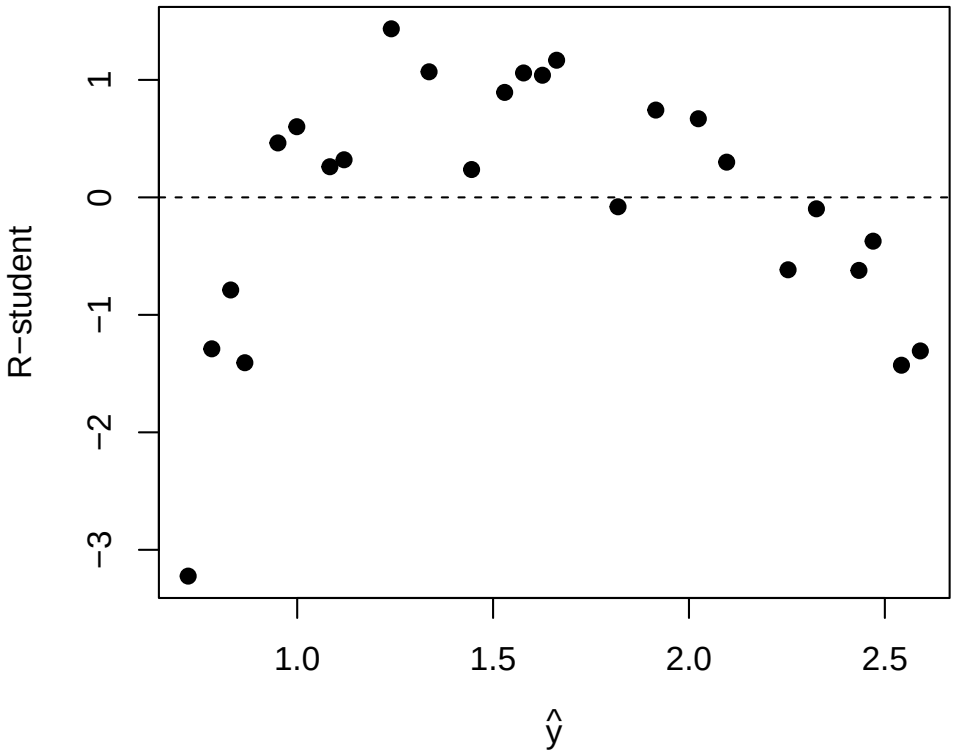
- y: DC output of a windmill, x: wind velocity (mph)
- $n = 25$ observations (MPV Table 5.5)
- Goal: model DC output as a function of wind velocity
- The scatter plot is clearly curved



Straight-Line Fit

$$\hat{y} = 0.1309 + 0.2411\,x$$
$$R^2 = 0.8745$$

- R-student against $\hat{y}$ shows a clear curved pattern
- The straight line is inadequate despite the high R^2



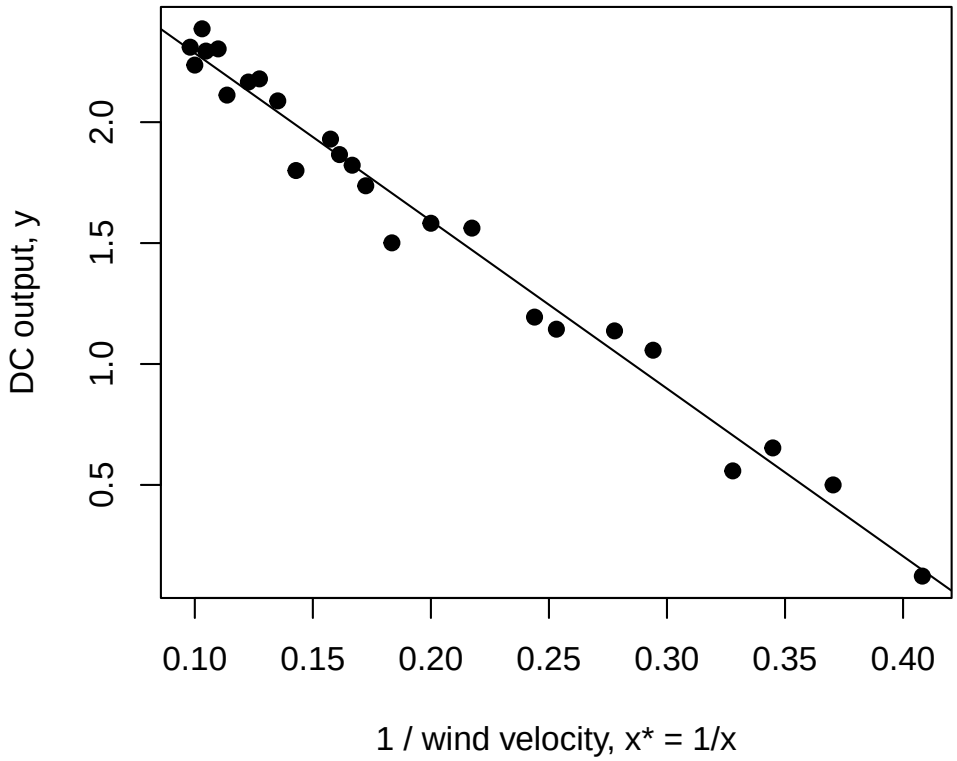
Question: Which Transformation?

DC output rises with wind velocity but levels off, and the spread around the trend looks roughly constant. Which transformation would you try, and on which variable?

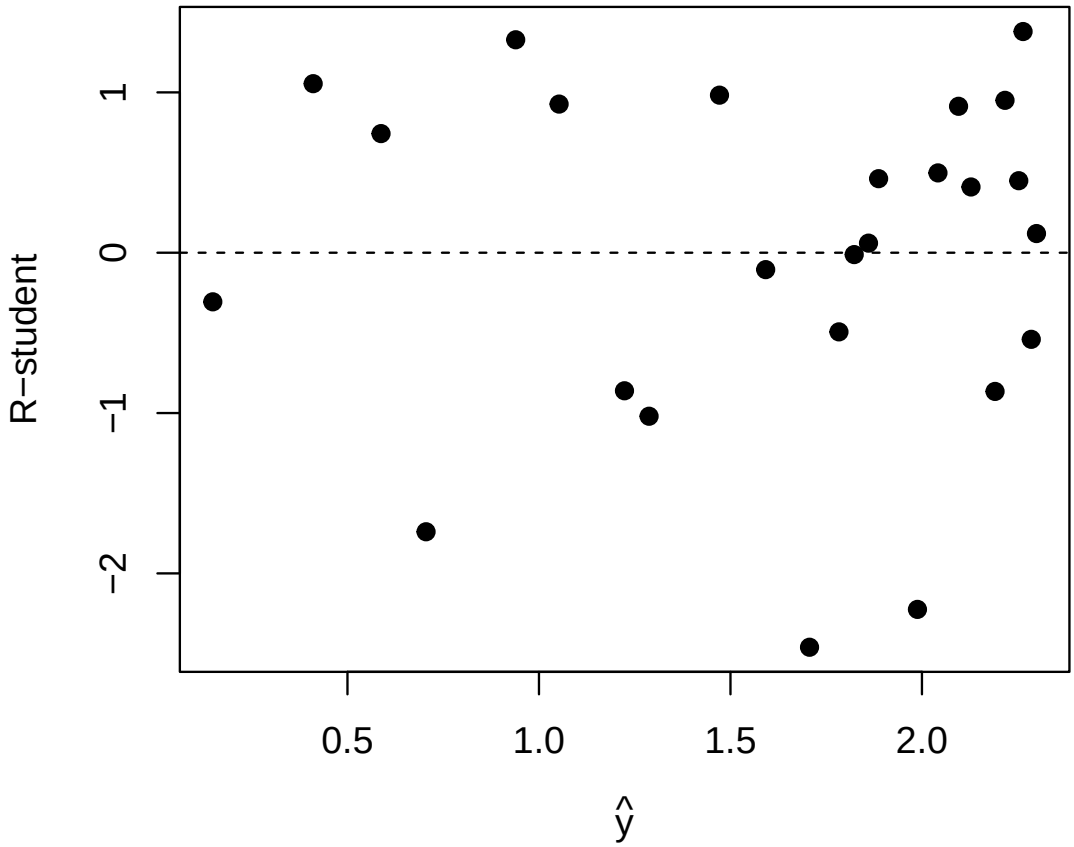
Reciprocal Transformation of x

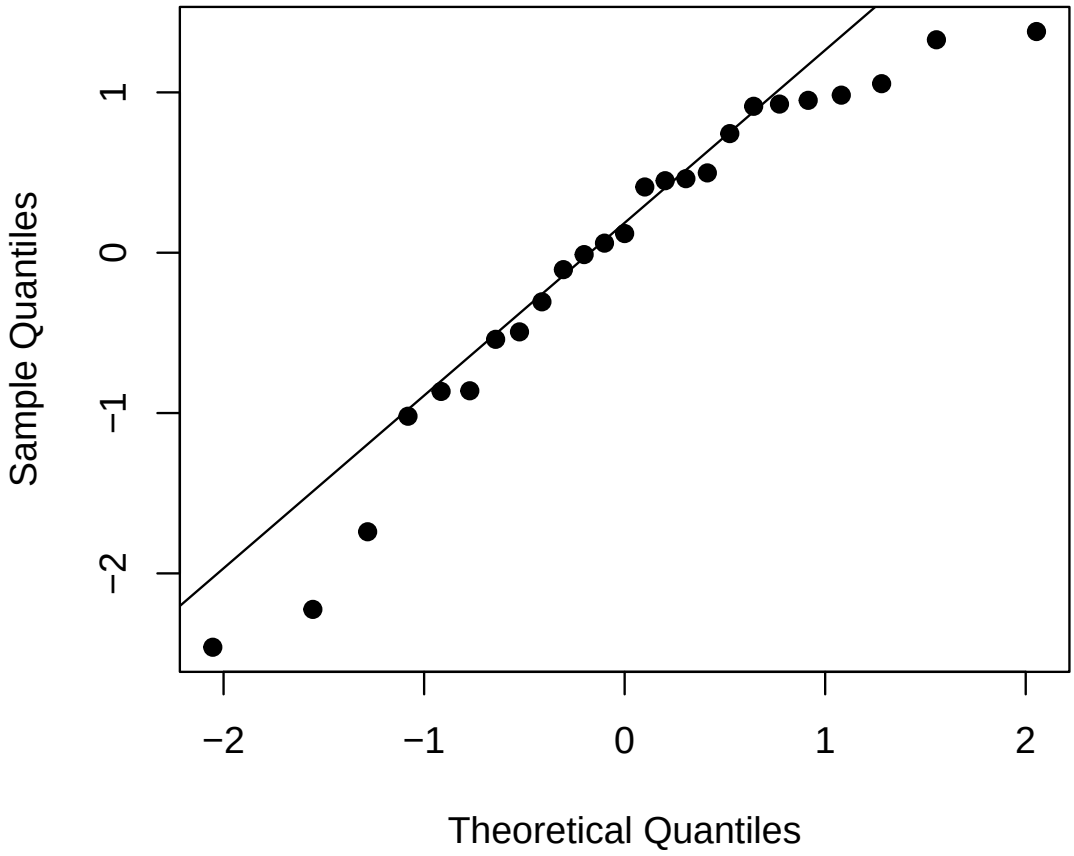
$$\begin{aligned} x^* &= 1/x \\ \hat{y} &= 2.9789 - 6.9345\, x^* \\ R^2 &= 0.9800 \end{aligned}$$

- The scatter plot against x^* is close to a straight line
- The fitted curve levels off toward 2.9789 as x grows, but the data stop at 10.2 mph, so this limit is an extrapolation



Residual Analysis After the Transformation


R−student
1
0
−1
−2
0.5
1.0
1.5
2.0
ŷ



- R-student against $\hat{y}$ shows no systematic pattern
- The normal probability plot is close to a straight line

6 The Box–Cox Method

Motivation for the Box–Cox Method

- The variance-stabilizing table requires knowing how σ^2 depends on $\mathrm{E}[y]$, which is usually unknown
- Box and Cox (1964): estimate the power of the transformation from the data
- Considers the power family y^λ, with λ estimated together with $\boldsymbol{\beta}$ by maximum likelihood
- Aims to correct nonnormality and nonconstant variance
- Requires $y_i > 0$; if some $y_i \leq 0$, shift to $y + c$ with c large enough, or use the Yeo–Johnson family, which allows any real y

The Power Transformation Family

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln y, & \lambda = 0 \end{cases}$$

- As $\lambda \to 0$, $(y^\lambda - 1)/\lambda \to \ln y$, so the logarithm belongs to the family
- $\lambda = 1$: no transformation, $\lambda = 1/2$: square root, $\lambda = -1$: reciprocal

Why $\mathrm{SS}_{\mathrm{Res}}$ Cannot Be Compared Across λ

- The units and size of $y^{(\lambda)}$ change with λ, so the raw $\mathrm{SS}_{\mathrm{Res}}$ changes with λ even when the fit does not
- Electric utility data, unscaled $\mathrm{SS}_{\mathrm{Res}}$:

Response	y	$\sqrt{y}$	$\ln y$
$\mathrm{SS}_{\mathrm{Res}}$	126.87	10.98	27.53

- $\sqrt{y}$ is not "ten times better" than y: its values are simply smaller

The Scaled Response

- Fix: divide by a factor involving the geometric mean $\dot{y}$, which puts every λ on the units of y

$$\tilde{y}^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda\, \dot{y}^{\lambda-1}}, & \lambda \neq 0 \\ \dot{y} \ln y, & \lambda = 0 \end{cases} \qquad \dot{y} = \exp\left[\frac{1}{n}\sum_{i=1}^{n} \ln y_i\right]$$

- $\tilde{y}^{(\lambda)} = y^{(\lambda)}/\dot{y}^{\lambda-1}$, so for each λ the two versions differ by a constant factor and give the same fitted model up to scale
- Check: at $\lambda = 1$, $\tilde{y}^{(1)} = y - 1$, so its $\mathrm{SS}_{\mathrm{Res}}$ is that of the straight-line fit to y

Model and Estimation

$$\tilde{\mathbf{y}}^{(\lambda)} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$$

- For fixed λ, $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\tilde{\mathbf{y}}^{(\lambda)}$
- Let $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ be the residual sum of squares from this fit

$$\mathrm{SS}_{\mathrm{Res}}(\lambda) = \sum_{i=1}^{n} \left(\tilde{y}_i^{(\lambda)} - \hat{\tilde{y}}_i^{(\lambda)}\right)^2$$

- With the scaled response, the maximum likelihood estimate $\hat{\lambda}$ is the value that minimizes $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ (derivation next)

Why Minimizing $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ Is Maximum Likelihood (1)

- Assume $y_i^{(\lambda)} = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \sim N(0, \sigma^2)$ independent (unscaled family)
- The density of y_i is the normal density of $y_i^{(\lambda)}$ times the Jacobian $\left|\mathrm{d}y_i^{(\lambda)}/\mathrm{d}y_i\right| = y_i^{\lambda-1}$

$$\ell(\boldsymbol{\beta}, \sigma^2, \lambda) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(y_i^{(\lambda)} - \mathbf{x}_i'\boldsymbol{\beta}\right)^2 + (\lambda - 1)\sum_{i=1}^{n}\ln y_i$$

- Without the Jacobian term, the likelihood would compare densities of different variables $y^{(\lambda)}$ rather than of the observed y

Why Minimizing $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ Is Maximum Likelihood (2)

- For fixed λ, maximize over $\boldsymbol{\beta}$ and σ^2: OLS gives $\hat{\boldsymbol{\beta}}$, and $\hat{\sigma}^2 = S(\lambda)/n$, where $S(\lambda)$ is the unscaled residual sum of squares
- Substituting gives the profile log-likelihood

$$\ell_p(\lambda) = \text{const} - \frac{n}{2}\ln S(\lambda) + (\lambda - 1)\sum_{i=1}^{n}\ln y_i$$

- Since $\sum_i \ln y_i = n \ln \dot{y}$, the last term is $\frac{n}{2}\ln \dot{y}^{\,2(\lambda-1)}$, so

$$\ell_p(\lambda) = \text{const} - \frac{n}{2}\ln\frac{S(\lambda)}{\dot{y}^{\,2(\lambda-1)}} = \text{const} - \frac{n}{2}\ln \mathrm{SS}_{\mathrm{Res}}(\lambda)$$

- Dividing $y^{(\lambda)}$ by $\dot{y}^{\lambda-1}$ divides $S(\lambda)$ by $\dot{y}^{2(\lambda-1)}$: the scaling absorbs the Jacobian

The Box–Cox Procedure

1. Choose a grid of values of λ, for example from -2 to 2
2. For each λ, compute the scaled $\tilde{y}^{(\lambda)}$, fit the model, and record $\mathrm{SS}_{\mathrm{Res}}(\lambda)$
3. Plot $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ against λ and locate the minimum $\hat{\lambda}$
4. Analyze the simple power y^{λ} (not the scaled version), using a convenient value of λ near $\hat{\lambda}$, and recheck the residuals

Later inference treats λ as known, so it is only approximate.

Confidence Interval for λ

$$\mathrm{SS}^* = \mathrm{SS}_{\mathrm{Res}}(\hat{\lambda})\left(1 + \frac{t^2_{\alpha/2,\,\nu}}{\nu}\right)$$

- ν is the residual degrees of freedom
- A $100(1-\alpha)\%$ confidence interval for λ contains all λ with $\mathrm{SS}_{\mathrm{Res}}(\lambda) \le \mathrm{SS}^*$
- If the interval contains 1, the data give no evidence that a transformation is needed
- Otherwise choose a convenient value inside the interval, such as -1, 0, $1/2$

Log-Likelihood Form and `MASS::boxcox`

- Up to a constant, the profile log-likelihood of λ is (as derived above)

$$L(\lambda) = -\frac{n}{2} \ln \mathrm{SS}_{\mathrm{Res}}(\lambda)$$

- The likelihood-based interval keeps all λ with $L(\hat{\lambda}) - L(\lambda) \leq \frac{1}{2}\chi^2_{\alpha,\,1}$
- Equivalently, $\mathrm{SS}_{\mathrm{Res}}(\lambda) \leq \mathrm{SS}_{\mathrm{Res}}(\hat{\lambda}) \exp(\chi^2_{\alpha,\,1}/n)$, which agrees closely with SS^* unless n is small
- `MASS::boxcox` plots $L(\lambda)$ with this 95% interval
- `car::powerTransform` reports $\hat{\lambda}$, a Wald interval, and likelihood-ratio tests of $\lambda = 1$ and $\lambda = 0$

7 Example: Box–Cox on the Electric Utility Data

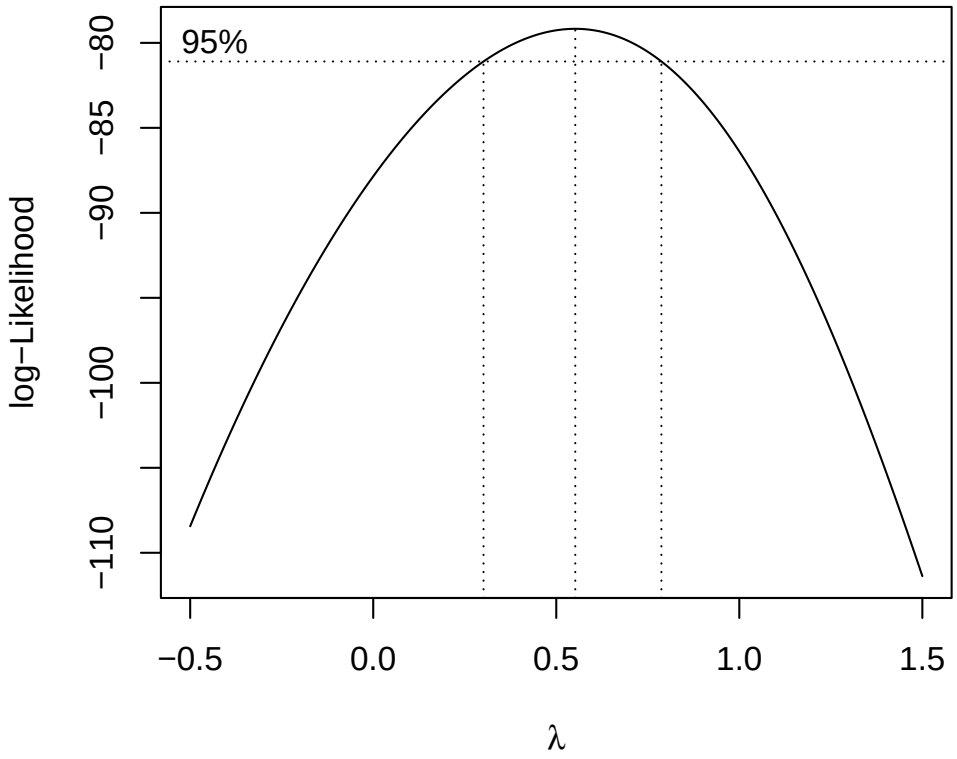
$\mathrm{SS}_{\mathrm{Res}}(\lambda)$ for the Electric Utility Data

λ	$\mathrm{SS}_{\mathrm{Res}}(\lambda)$
-1	986.03
-0.5	291.58
0	134.09
0.25	107.21
0.5	96.95
0.75	101.69
1	126.87

- The scaled $\mathrm{SS}_{\mathrm{Res}}(\lambda)$ is smallest near $\lambda = 0.5$; the minimum is $\mathrm{SS}_{\mathrm{Res}}(0.552) = 96.63$
- $\mathrm{SS}^* = 96.63\,(1 + 2.008^2/51) = 104.27$: $\lambda = 0.5$ and 0.75 lie inside the interval, 0.25 and 1 outside
- $\mathrm{SS}_{\mathrm{Res}}(1) = 126.87$ equals the $\mathrm{SS}_{\mathrm{Res}}$ of the straight-line fit to y

Box–Cox Plot from `MASS`

- $\hat{\lambda} = 0.55$
- 95% interval: $(0.30,\ 0.79)$
- Contains $\lambda = 0.5$
- Excludes $\lambda = 1$ and $\lambda = 0$



Question: Which λ Should We Use?

The estimate is $\widehat{\lambda} = 0.55$ with 95% interval $(0.30, 0.79)$. Which value would you use for the final model, and why?

- Use $\lambda = 0.5$, that is $\sqrt{y}$: it lies inside the interval and is easy to interpret
- It agrees with the transformation chosen from the residual plots and the variance-stabilizing table
- The logarithm ($\lambda = 0$) and no transformation ($\lambda = 1$) are both outside the interval

8 The Box–Tidwell Method

Motivation for the Box–Tidwell Method

- Box–Cox transforms y; when the problem is nonlinearity with constant variance, transform the regressor instead
- Model, with unknown power α:

$$y = \beta_0 + \beta_1 \xi + \varepsilon, \qquad \xi = \begin{cases} x^\alpha, & \alpha \neq 0 \\ \ln x, & \alpha = 0 \end{cases}$$

- Box and Tidwell (1962) estimate α iteratively
- Special case: $\alpha = 0$: logarithm

Linearizing the Model in α: Setup

- The model $\mathrm{E}[y] = \beta_0 + \beta_1 x^\alpha$ is nonlinear in the unknown power α
- Idea: replace it by a first-order Taylor series in α about a starting value α_0, treating x, β_0, β_1 as fixed

$$f(\alpha) \approx f(\alpha_0) + (\alpha - \alpha_0)\, f'(\alpha_0), \qquad f(\alpha) = \beta_0 + \beta_1 x^\alpha$$

- Take $\alpha_0 = 1$: then $x^{\alpha_0} = x$, the untransformed straight-line model
- We need $f'(\alpha)$, which requires $\partial x^\alpha / \partial \alpha$

Step 1: Differentiate x^α with Respect to α

- Here x is a constant and α is the variable, so x^α is an exponential function of α, not a power function of x
- Rewrite it with base e (this requires $x > 0$)

$$x^\alpha = e^{\alpha \ln x}$$

- $\ln x$ is a constant with respect to α, so the chain rule gives

$$\frac{\partial}{\partial \alpha}\, x^\alpha = \frac{\partial}{\partial \alpha}\, e^{\alpha \ln x} = (\ln x)\, e^{\alpha \ln x} = x^\alpha \ln x$$

- Do not confuse this with $\partial x^\alpha / \partial x = \alpha x^{\alpha - 1}$, which differentiates with respect to x

Step 2: Evaluate the Taylor Expansion at $\alpha_0 = 1$

β_0 and β_1 do not depend on α, so differentiating $f(\alpha) = \beta_0 + \beta_1 x^\alpha$ gives

$$f'(\alpha) = \beta_1 \frac{\partial x^\alpha}{\partial \alpha} = \beta_1\, x^\alpha \ln x$$

Evaluate f and f' at $\alpha_0 = 1$:

$$f(1) = \beta_0 + \beta_1 x, \qquad f'(1) = \beta_1\, x \ln x$$

Substitute into $f(\alpha) \approx f(1) + (\alpha - 1)\, f'(1)$:

$$\mathrm{E}[y] \approx \beta_0 + \beta_1 x + (\alpha - 1)\, \beta_1\, x \ln x$$

Step 3: Reparameterize to a Linear Model

- $(\alpha - 1)\beta_1$ is a single unknown constant, and $x \ln x$ can be computed from the data

$$\gamma = (\alpha - 1)\beta_1, \qquad w = x \ln x$$

$$\mathrm{E}[y] \approx \beta_0 + \beta_1 x + \gamma w$$

- The model is linear in the regressors x and w, so the coefficients can be estimated by OLS
- Recover the power by solving $\gamma = (\alpha - 1)\beta_1$ for α: $\alpha = \gamma / \beta_1 + 1$
- $\gamma = 0$ exactly when $\alpha = 1$, that is, when no transformation of x is needed

One Iteration of the Procedure

1. Fit $y = \beta_0 + \beta_1 x + \varepsilon$ and record $\hat{\beta}_1$
2. Fit $y = \beta_0^* + \beta_1^* x + \gamma w + \varepsilon$ with $w = x \ln x$ and record $\hat{\gamma}$
3. Compute the first estimate of the power

$$\hat{\alpha}_1 = \frac{\hat{\gamma}}{\hat{\beta}_1} + 1$$

- The coefficients in step 2 are starred because the linearized model holds only approximately, so $\hat{\beta}_0^*$, $\hat{\beta}_1^*$ need not estimate β_0, β_1 well
- Following Box and Tidwell, $\hat{\beta}_1$ is taken from the fit in step 1, not from step 2

Iterating

- Set $x^{(1)} = x^{\hat{\alpha}_1}$ and repeat the procedure with $x^{(1)}$ in place of x
- Stop when $\hat{\alpha}_k \approx 1$, that is, when no further transformation is needed
- The overall estimate is the product $\hat{\alpha} = \prod_k \hat{\alpha}_k$, since a power of a power multiplies the exponents
- The t statistic for $\hat{\gamma}$ gives an approximate test of $H_0 \colon \alpha = 1$
- In R, `car::boxTidwell` runs the iterations and reports $\hat{\alpha}$ with this test

9 Example: Box–Tidwell on the Windmill Data

Windmill Data: First Iteration

$$\hat{\beta}_1 = 0.2411, \qquad \hat{\gamma} = -0.4626 \quad (t = -9.13)$$

$$\hat{\alpha}_1 = \frac{-0.4626}{0.2411} + 1 = -0.918$$

- The significant $\hat{\gamma}$ confirms that the straight line is inadequate
- $\hat{\alpha}_1$ is close to -1, which suggests the reciprocal transformation

Windmill Data: Iterations

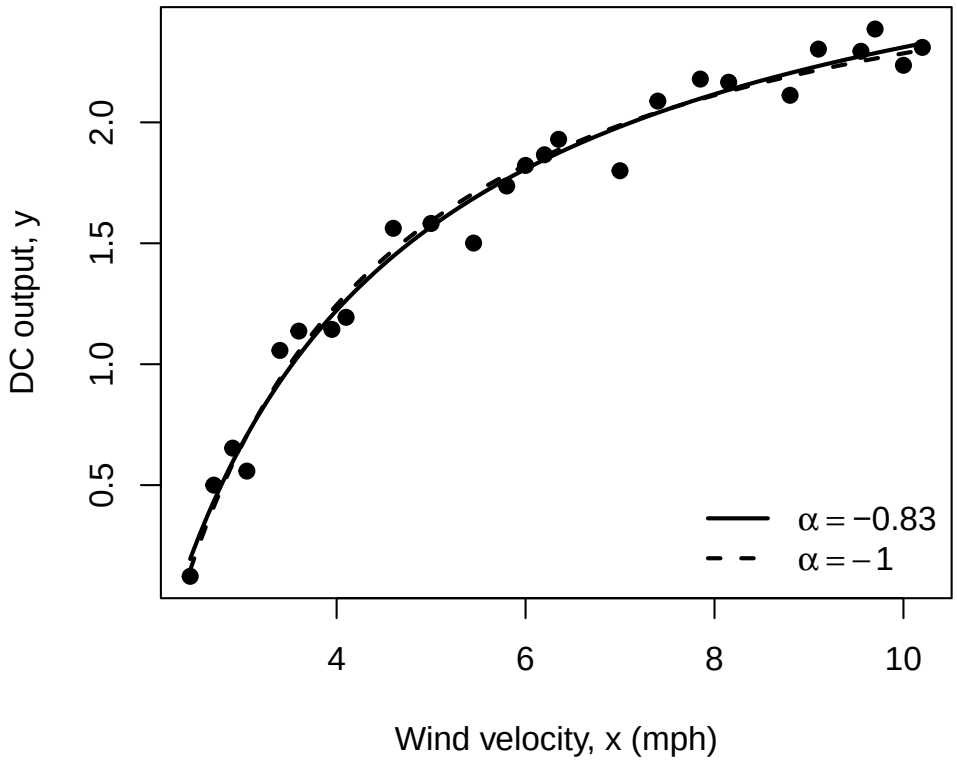
k	$\hat{\beta}_1$	$\hat{\gamma}$	$t(\hat{\gamma})$	$\hat{\alpha}_k$	cumulative $\hat{\alpha}$
1	0.2411	−0.4626	−9.13	−0.9183	−0.9183
2	−6.6784	0.5994	0.51	0.9103	−0.8359
3	−6.4738	0.0190	0.02	0.9971	−0.8334
4	−6.4686	0.0006	0.00	0.9999	−0.8333

- From the second iteration on, $\hat{\gamma}$ is not significant and $\hat{\alpha}_k \approx 1$
- The procedure converges to $\hat{\alpha} = -0.833$

Box–Tidwell Fit Compared with the Reciprocal Model

$$\hat{y} = 3.2601 - 6.4684\, x^{-0.833}$$

	$\alpha = -0.833$	$\alpha = -1$
R^2	0.9809	0.9800
$\hat{\sigma}$	0.0920	0.0942



Question: $x^{-0.833}$ **or** $1/x$**?**

The estimated power is -0.833, but the reciprocal model has almost the same R^2. Which model would you report?

- The reciprocal model: R^2 and $\hat{\sigma}$ are essentially unchanged and the fitted curves nearly coincide
- $1/x$ is simpler to interpret: the intercept is the level the fitted curve approaches as x grows
- Treat that level with care: the two fits agree within 2.45–10.2 mph, yet their limits are 2.98 and 3.26, so the data barely determine the upper limit

10 Summary

Summary

- Nonconstant variance: transform y using the variance-stabilizing table, then recheck the residuals
- Unknown variance–mean relationship: estimate the power of y with the Box–Cox method
- Curved relationship with constant variance: transform x, guided by the shape of the scatter plot
- Unknown power for x: estimate it with the Box–Tidwell method
- After any transformation, repeat the residual analysis and interpret results on the transformed scale
- Back-transformed predictions estimate the median of y; log-scale coefficients are percentage changes